VOL. I WEEKDAY EDITION

THE ELON GAZETTE

Weekday briefing on what Elon Musk is saying and doing

TUESDAY, NOVEMBER 10, 2026

THE MORNING PICTURE

Grok 4.8 nears a soft launch on code borrowed from Starlink

Elon Musk said early Tuesday that pre-training of Grok 4.8 is finished, that the model is now in reinforcement learning, and that a soft rollout to Grok Bot users should follow later this week. To get there, he said, xAI threw out its JAX and SGLang software and rewrote its training and inference stack from scratch in C and C++, using work done by the engineers who built Starlink. On Monday he marked the start of site work on Tesla's $10 billion "Project Crystal Sun" solar factory near Houston, highlighted a SpaceX executive's claim that Grok spots debris in rocket parts better than human inspectors, and said "Gwynne is great" about SpaceX President Gwynne Shotwell.

I · XAI & GROK

"This time, the foundations are built right"

Grok 4.8 has finished pre-training and is in reinforcement learning on a stack xAI rewrote in C and C++; Musk says Grok 5 could pass Anthropic's best model in two or three months.

RL	This week	2–3 mo
Stage Grok 4.8 is in now, with pre-training done	Planned soft rollout of Grok 4.8 to Grok Bot users	When Musk said Grok 5 might exceed Opus 5.5

Asked on X when to expect Grok 4.8, Musk answered at 4:35 a.m. EST Tuesday: "We have dumped JAX/SGLang and rewritten the entire RL and inference stack from scratch in C/C++, which took some extra time. Training (aka pre-training) is done. Lot of cleanup was needed on the SFT data to get that mostly right. Now, 4.8 is in RL and the evals are starting to climb. We will do a soft rollout to Bot users later this week and see how it goes. Grok 4.8 will bear no resemblance to 4.7 (thankfully). It is not yet incredible, but, this time, the foundations are built right, so I expect it to evolve to be something special."

SFT is supervised fine-tuning, training on curated example answers; RL, reinforcement learning, is the later stage in which the model is trained against rewards. JAX is a Google-originated Python library for accelerator computing and SGLang an open-source serving framework. FourWeekMBA noted that the update came 57 days after Musk wrote on September 14 that Grok 4.8, "a 2.5T model trained with our new C++ software stack," would finish training that week. No benchmark scores have been published for the new model.

C, a little C++ and some assembly

When programmer tetsuo (@tetsuoai) cheered "C wins again," Musk replied at 4:46 a.m. EST: "C with a little C++, but not too much." Asked by another user why the stack was not written in Rust, he answered: "Real men manage their own memory." At 5:14 a.m. he added: "At SpaceX, we write our own headers and calculate our own time (includes leap seconds). Not many people will know what this means, but those that do will appreciate it. The new Grok AI/SI training and inference stack is derived from incredible of work done by amazing engineers to create Starlink." Three minutes later he wrote: "(There's even some assembly). One day, super intelligence will write a better software stack, but that day has not yet come. This is tight code."

> *"Turns out that the Starlink software, built for managing the movement of vast amounts of data through space and to the ground, is also very well suited to training super intelligence"*
>
> ELON MUSK · 5:34 A.M. EST TUESDAY

Grok 5 against Opus 5.5

Asked how Grok 4.8 will compare with Anthropic's Claude Opus 5.5, Musk replied at 5:22 a.m. EST: "Not as good. Opus 5.5 is a masterpiece. But Grok 5 should exceed Opus 5.5 and maybe that happens in 2 or 3 months, with continued acceleration thereafter."

"The most entertaining outcome"

At 9:49 p.m. EST Monday, Tesla investor and engineer James Douma wrote that one "irony maximizing outcome" he was not looking forward to "is Anthropic proving to the world that their fears are justified by building the super-intelligence that they fear." Musk quoted him 14 minutes later: "The most entertaining outcome is the most likely, especially if ironic."

II · SPACEX

Grok finds the debris human inspectors miss

SpaceX's head of launch says AI now runs through every level of the company, and that Falcon 9 is on its way to retirement.

99.9%	94%	20×
Grok's accuracy finding foreign object debris in borescope video	Human inspectors' accuracy on the same task	Bandwidth a Florida Starship launch can carry versus one Falcon 9

At 10:04 p.m. EST Monday Musk quote-posted a single word, "Grok," over a post by Sawyer Merritt on remarks by Kiko Dontchev, SpaceX's vice president of launch. Speaking at last week's UP.Summit conference in Florida, Dontchev said the job of finding foreign object debris, or FOD, which can cause catastrophic damage to engines, has been largely automated with Grok, Ars Technica reported. "We'll have an inspector go in and borescope a tube, and we find that humans are about 94 percent perfect (at) finding a piece of it. Grok analyzing the same video is 99.9 percent perfect," he said. SpaceX also uses AI for visual inspection of Starlink customer terminals.

"In the last couple years, when I talk about SpaceX, yes, we are a rocket company, but fundamentally we are an infrastructure company at this point," Dontchev said, adding that "every single person, including the technicians and the technician leaders," now uses AI day to day. Tasks that took hours or days now take seconds or minutes, he said.

Falcon 9 fades, Pad 39A is ready

Dontchev said Falcon 9 must give way because its second stage is still thrown away on every flight, while Starship is designed to have both stages caught. A Florida Starship launch could carry up to 20 times the bandwidth of a single Falcon 9, he said, and "Pad 39A is effectively complete." SpaceX hopes to start moving boosters and ships to the Cape by barge in December "and then maybe even fly by the end of the year," Ars Technica reported.

ON THE WIRE

Starlink code in Grok. Musk wrote that the new Grok training and inference stack is derived from work done by the engineers who built Starlink (post).



Launch Complex 39A at Kennedy Space Center, Florida, on Jan. 18, 2026: the Falcon pad's service structure at left, the Starship launch tower in the middle and a SpaceX hangar at right.

PHOTO: OLGA ERNST / WIKIMEDIA COMMONS · CC BY-SA 4.0 (CROPPED)

QUICK HIT

BT holds Starlink. BT chief executive Allison Kirkby said its Starlink-based rural broadband service is "ready to launch" but will wait until "the noise has died down a little bit" over last month's Starlink limits in Ukraine, The Telegraph reported.

"Gwynne is great"

At 4:47 p.m. EST Monday Musk wrote "Gwynne is great" over a clip of SpaceX President and Chief Operating Officer Gwynne Shotwell saying, "That's the man I've worked for for 24 years. I love him." The remark comes from her IPO-day interview on June 12, when she told CNBC that Musk is "very misunderstood across the board" and that he "wanted regular people to be able to buy the stock."

ON THE WIRE

Terminal checks. Beyond engine parts, SpaceX also uses AI to visually inspect the Starlink customer terminals it builds, Ars Technica reported.

Site work starts on "Project Crystal Sun"

Tesla's planned $10.1 billion solar plant in Fort Bend County, Texas, would make everything from silicon ingots to finished modules.

$10.1B	Q1 2029	9,712
Planned investment, including about $8.6B in equipment	Target start of commercial operations	Permanent jobs Tesla projects by 2033

At 10:21 a.m. EST Monday Musk quote-posted "Project Crystal Sun" over a post by Sawyer Merritt saying Tesla had started construction of the solar factory near Houston, with construction expected to finish in 2028. Drone footage from videographer S.E. Robinson, Jr. shows crews clearing and leveling the roughly 3,000-acre site in Fort Bend County, along with equipment staging areas, Teslarati reported; Tesla has not issued a formal groundbreaking announcement.

Filings made in July under Texas's Jobs, Energy, Technology and Innovation Act describe a vertically integrated plant covering silicon ingots, wafers, coated cells and finished modules. Tesla projects about 1,900 jobs in the first year of operations, rising to 9,712 by 2033, plus a construction peak of about 1,147 workers; an accompanying analysis estimates the project could add about $107 billion to Texas's economic output over several decades. Lamar Consolidated Independent School District approved school-tax limits in September, and Fort Bend County commissioners approved abatement agreements in late October. Of the $10.1 billion, about $8.6 billion is for equipment and $1.5 billion for real property, Tesla North reported, citing the school district's summary.

AI6: less memory, same bandwidth

When Tesla investor Herbert Ong shared a report that Tesla had cut the LPDDR6 memory on its next-generation AI6 chip for Optimus from 216GB to 144GB, Musk replied at 10:20 a.m. EST Monday: "Memory bandwidth was held constant though." He first announced the cut on October 1, saying, "This was the only way to get enough volume for Optimus production and greatly reduces cost," according to Tesla North.

> **QUICK HIT**
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> **APEC trip.** Musk is expected to join President Trump's delegation of US business leaders to next week's APEC summit, set for November 18–19 in Shenzhen, Reuters reported, citing the South China Morning Post.

Cybercab and the ride-hailing bet

Software founder Carl Hancock asked how turning Tesla into a taxi company makes sense when the five largest ride-hailing companies have about 16 million drivers and a combined market value of $175 billion. Palmer Luckey, founder of Anduril and Oculus, replied: "Take that ridesharing revenue ($150B+) and subtract the cost of paying 16,000,000 drivers. That is the bet." At 4:44 a.m. EST Tuesday Musk added: "Cybercab will do a majority of all passenger miles in any jurisdiction where it is not prohibited by law."

Who founded Tesla

Musk replied "Exactly" at 4:13 a.m. EST Tuesday to a long post by investor Captain Eli recounting that Musk supplied $6.5 million of Tesla's $7.5 million Series A in February 2004 and became chairman, and that the 2009 settlement of Martin Eberhard's lawsuit named five co-founders: Eberhard, Marc Tarpenning, Musk, JB Straubel and Ian Wright.

Takeaways

1. Grok 4.8 has finished pre-training and is in reinforcement learning; a soft rollout to Grok Bot users is planned for later this week.
2. xAI rewrote its training and inference stack in C and C++, drawing on Starlink software, Musk said.
3. Musk said Grok 5 might exceed Claude Opus 5.5 in two or three months.
4. Site work has begun on Tesla's $10.1 billion Project Crystal Sun solar factory near Houston.
5. SpaceX says Grok finds debris in rocket parts with 99.9% accuracy, against 94% for human inspectors.

Sources

Posts by @elonmusk, @cradrrat, @tetsuoai, @Owl_of_Atena, @Dimdv99, @jamesdouma, @SawyerMerritt, @teslaownersSV, @herbertong, @carlhancock, @PalmerLuckey and @TheCaptainEli.

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